Grade: Check Plus

2. Expected Value and Options

x	$f(x)$	μ	σ^2	v	$g(v) = P(V=v$
20	.05	1	1.46	0	
21	.06	1.26	1.17	6	
22	.07	1.54	0.81	0	
23	.08	1.84	0.46	0	
24	.09	2.16	0.18	0	
25	.12	3	0.02	0	.47
26	.15	3.9	0.05	1	.15
27	.13	3.51	0.33	2	.13
28	.10	2.8	0.67	3	.10
29	.10	2.9	1.29	4	.10
30	.05	1.5	1.05	5	.05
	1.00	25.41	7.5		1.00

1) $\mu = 25.41$ ✓Q1 $\quad \mu = x\, f(x)$

$\sigma = 2.74$ $\quad \sigma^2 = \sum (x-\mu)^2 f(x)$

$\sigma = \sqrt{\sigma^2}$

2a) $\mu - \sigma = 22.67$ $\quad 22.67 \le x \; 28.15$

$\mu + \sigma = 28.15$ $\quad 23 \le x \; 28$

23, 24, 25, 26, 27, 28

.08 + .09 + .12 + .15 + .13 + .10 = .67

✓Q2a

* Probability = 67% within 1 standard div

2b) $\mu - 2\sigma = 19.93$

$\mu + 2\sigma = 30.89$

$19.93 \le x \le 30.89$

$20 \le x \le 30$

all values are within this range

✓Q2b

* Probability = 100% within 2 std div

3)

Value (v)	g(v)
0	.47
1	.15
2	.13
3	.10
4	.10
5	.05

✓ Q3

4) Probability $V > 0$

✓ Q4

$.15 + .13 + .10 + .10 + .05 = .53$

Prob = 53%

5)

(v)	g(v)	μ	σ^2
0	.47	0	.87
1	.15	.15	.02
2	.13	.26	.05
3	.10	.30	.27
4	.10	.40	.70
5	.05	.28	.66
		1.36	2.57

$\mu = 1.36$

✓ Q5

$\sigma = \sqrt{2.57} = 1.60$